

7. The article below was taken from a newspaper in early 2017.

“DNA profiling snares Welsh sheep rustler after investigation.”

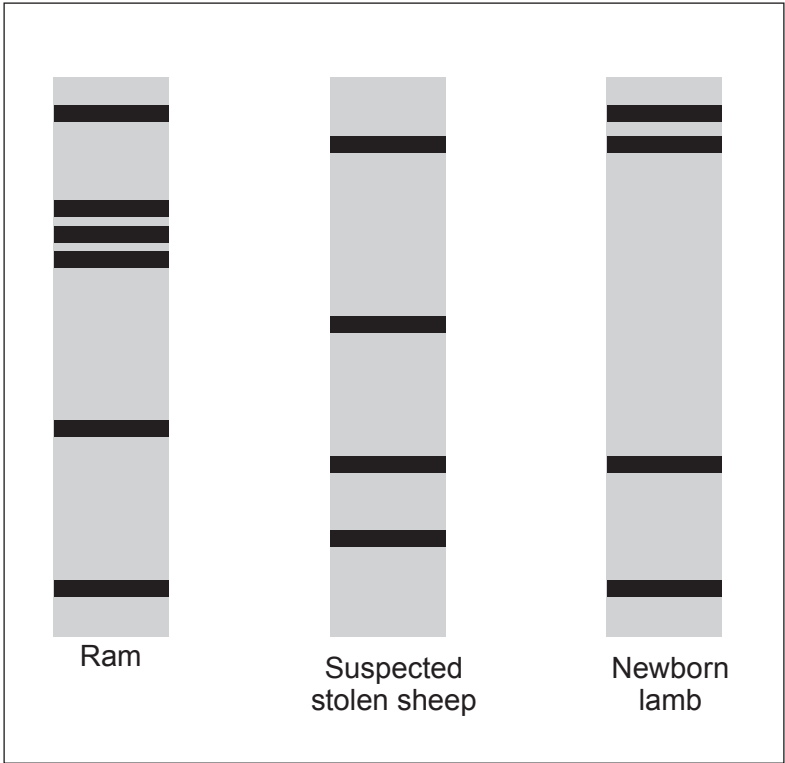
Daily Post 4/4/2017

In January 2015, a number of pregnant sheep were stolen from a farm. Enquiries within the local farming community revealed a pregnant sheep, which was suspected of having been stolen, had been sold at a livestock market.

The Animal Plant Health Agency (APHA) wanted to carry out DNA analysis on the sheep and its lamb. They had to wait two months for the sheep to give birth before taking a sample of blood from the lamb.

The diagram below shows a genetic profile of:

- the ram (known to be the father of the lamb in the suspected stolen sheep)
- the suspected stolen sheep
- the newborn lamb.



Examiner
only

(a) Describe the process of genetic profiling. [2]

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(b) Explain how the genetic profile provides evidence that the sheep had been stolen. [2]

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(c) State **one** possible health benefit and suggest **one** ethical concern linked to DNA profiling in humans. [2]

Possible health benefit

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Ethical concern

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